

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458824.6376 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.078 +/- 0.05
Transit depth (Rp/Rs)^2	0.61 +/- 0.78 [%]
Orbital Inclination (inc)	84.38 +/- 1.69
Airmass coefficient 1 (a1)	0.78 +/- 0.58
Airmass coefficient 2 (a2)	0.32 +/- 0.77
Scatter in the residuals of the lightcurve fit is	83.84 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	3.0
Transit Duration (day)	0.024 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.078 ± 0.05 vs 0.131 (deviation 1.06σ)
Duration fit vs published	0.024 d vs 0.1075 d (deviation 2.53σ)
Mid-time O–C	4.3 min
Depth SNR	1.6
Residual scatter	83.84 %

Light curve

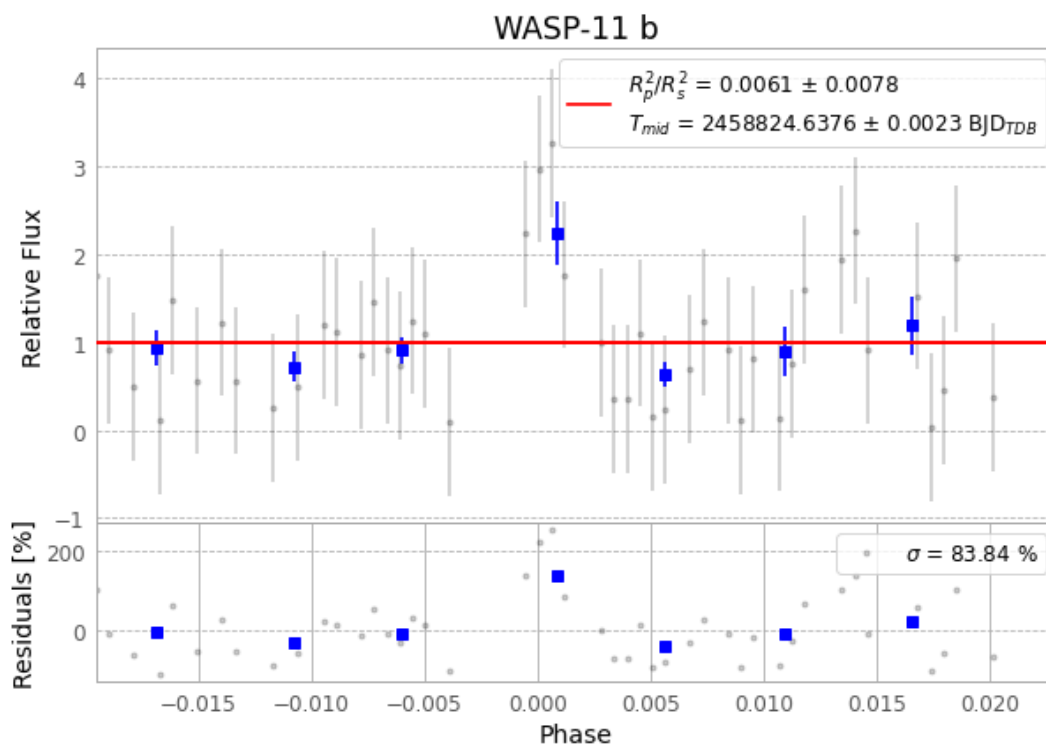


Figure 1 — FinalLightCurve_WASP-11 b_2019-12-06.png (SHA-256 11e95dda169a082f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [497, 421], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e6a183f6fc90ab90...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

76 FITS frames (50 MB), HATP-10191207012412.FITS → HATP-10191207051512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-191207.log (5645ee875656...), FinalLightCurve_WASP-11 b_2019-12-06.png (11e95dda169a...), FinalLightCurve_WASP-11 b_2019-12-06.csv (4986b1afecb5...)

Compute resources

Wall time 190.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 22:21Z.